Darius Muglich

Machine learning researcher · DPhil student, University of Oxford

dariusmuglich.com / Google Scholar

Education

University of Oxford 2023–present

DPhil · Autonomous Intelligent Machines and Systems (AIMS CDT)

Supervisor: Jakob Foerster

Thesis on policy selection in reinforcement learning: how learning biases and structural constraints shape learned behaviour and generalization.

Current work on logic gate networks and neurosymbolic robot control.

University of Oxford · MSc, Advanced Computer Science 2020–2021

University of British Columbia 2015–2020

BSc · Combined Honours in Mathematics and Computer Science

Research and industry experience

UC Berkeley · Center for Human-Compatible AI Jun–Sep 2025

Visiting Researcher

Learned latent representations of human strategies from unlabeled demonstrations, enabling the construction of diverse AI partners for evaluating human–AI coordination.

Bazaarvoice Aug 2022–Sep 2026

Data Scientist V · Full-time initially; part-time during the doctorate

Primary engineer for a bandit-based campaign optimization method, delivering an estimated 40% lift in incremental revenue over an analyst-designed heuristic in a multi-month company A/B test.

Selected publications

Ad-Hoc Teamwork from Human Demonstrations NeurIPS 2026

Darius Muglich, Niklas Lauffer, Tin Dizdarevic, Jakob Foerster

Learned latent human conventions from unlabeled demonstrations and constructed diverse proxy partners, revealing sensitivity of agent rankings to evaluation-partner coverage.

Expected Return Symmetries ICLR 2025

Darius Muglich, Johannes Forkel, Elise van der Pol, Jakob Foerster

Learned symmetries that preserve expected return, enabling coordination between independently trained agents without known environment symmetries.

Equivariant Networks for Zero-Shot Coordination NeurIPS 2022

Darius Muglich, Christian Schroeder de Witt, Elise van der Pol, Shimon Whiteson, Jakob Foerster

Developed symmetry-aware architectures and a test-time policy transformation to improve coordination between independently trained agents.

Generalized Beliefs for Cooperative AI ICML 2022

Darius Muglich, Luisa Zintgraf, Christian Schroeder de Witt, Shimon Whiteson, Jakob Foerster

Trained autoregressive transformers to infer hidden state from interaction histories and generalize to unfamiliar partner conventions, improving cooperative play.

Selected awards

Toshiba Research and EPSRC CDT Studentship · Fully funded doctoral studentship

NSERC Undergraduate Student Research Award · Three awards